

**PRISM: A MULTIMODAL DEEP LEARNING SYSTEM FOR PROGRESSIVE REAL-
TIME IDENTIFICATION OF SYNTHETIC MEDIA AND DISINFORMATION
ON SOCIAL MEDIA PLATFORMS**

Research Paper
Submitted to the Panel of Judges at
Mapua University - Makati Campus
City of Makati, Metro Manila

In partial fulfillment
of the requirements for the
Next Gen Start-up Competition 2026

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May 2026

INTRODUCTION

In the Philippines, often dubbed the "Social Media Capital of the World," the line between reality and fabrication is being blurred at an unprecedented speed, turning digital feeds into a battlefield where seeing is no longer believing. The meteoric rise of generative Artificial Intelligence (AI) tools has made it trivially easy to produce fabricated images, deepfake videos, and coordinated disinformation at an industrial scale. Jha (2025) highlights that social media fraud has emerged as a critical cybersecurity challenge, causing over USD 1.2 billion in reported losses in 2023 alone, where existing rule-based detection methods fundamentally fail to address the dynamic and multimodal nature of these scams. This threat is not abstract for Filipino users: Ong and Cabañes (2018) documented that the Philippines has already served as a testing ground for organized disinformation networks, where troll farms and fake news producers operate within deliberate hierarchical structures designed to manipulate public opinion at scale. More recently, Schipper (2025) warned that in the lead-up to the 2025 Philippine elections, generative AI tools were actively weaponized to spread revisionist narratives, deepen algorithmic bias, and undermine democratic participation, underscoring that the threat has only escalated since.

The feasibility of a locally-deployed detection solution has already been demonstrated. In May 2025, the Department of Information and Communications Technology (DICT) recognized Tunai, a Facebook-embedded AI fact-checking tool built by a Filipino development team, as the winner of the inaugural OpenGov Hackathon. DICT Secretary Henry Aguda noted that the tool demonstrated that Filipino developers can build technologies that uphold the truth and strengthen the connection between government and citizens (DICT, 2025). While Tunai addresses text-based fake news within a single platform, it does not extend to image forensics, deepfake

video detection, or browser-based passive detection, gaps that PRISM is specifically designed to fill.

Effective disinformation detection, however, requires far more than simple text analysis. Visweswaran et al. (2024) establish that unlike traditional news articles that predominantly rely on textual content, multimodal fake news leverages the persuasive power of visual elements, which necessitates the fusion of text and image modalities using technologies like Vision Transformers (ViT) to increase accuracy and reliability. Jing et al. (2023) further demonstrate that a progressive fusion framework that will capture representational information at multiple levels achieves at least 4.3% higher accuracy on real-world datasets compared to prior state-of-the-art methods, ensuring higher classification precision and adaptability across diverse content types in real-time applications. Pathak et al. (2023) add a critical platform-level dimension, showing that popularity-based and network-based recommendation algorithms on social networks disproportionately amplify misinformation, meaning that detection must keep pace with the speed at which false content will be surfaced to users algorithmically.

The deepfake dimension of the problem is particularly acute in the Philippine context. Blancaflor et al. (2023) found that Filipino awareness of deepfakes correlates strongly with social media exposure, yet legislative and technical safeguards remain critically underdeveloped. A companion study by Blancaflor et al. (2024) further revealed that deepfake technology is increasingly weaponized for blackmail and non-consensual intimate imagery targeting young Filipinos, reinforcing the urgency for accessible, real-time detection tools. On the technical side, the challenge is significant: Heidari et al. (2023) conducted a comprehensive review establishing that Convolutional Neural Networks (CNN) are the dominant methodology for deepfake detection, while a subsequent survey by Akhtar et al. (2024) classified the specific challenges of deepfake video detection into data, training, and reliability dimensions, noting that real-time deployment remains an open research problem. For AI-generated images, Park et al. (2024) demonstrated that artifact-based detection methods perform

well on GAN-generated images while image-encoder-based methods excel on diffusion model outputs, implying that a robust system must address both generation paradigms. Baraheem and Nguyen (2023) confirmed that CNN-based transfer learning frameworks can achieve high accuracy in distinguishing GAN-generated from real images, providing the foundational approach adopted in this study.

A significant gap remains in localized media forensics. Cruz et al. (2020) established that low-resource languages like Filipino lack the large labeled datasets that most detection systems assume, and demonstrated that multitask transfer learning can achieve 96% accuracy on the first expertly curated Filipino fake news benchmark dataset. This work directly motivates PRISM's approach of fine-tuning pre-trained multilingual models on localized Taglish data rather than training from scratch. Diwan et al. (2021) further argue that many existing NLP systems lack comprehensiveness regarding language specificity across diverse low-resource languages, reinforcing the need for a Taglish-aware detection module. Meanwhile, Valdez et al. (2021) documented from a Philippine classroom perspective that the spread of misinformation will be compounded by a lack of critical evaluation skills among Filipino internet users, highlighting that a detection tool must also serve an educational function by being transparent about its reasoning.

This transparency requirement is precisely where explainability becomes non-negotiable. Szczepański et al. (2021) demonstrate that high model performance alone is no longer sufficient for real-life scenarios; the explainability of the system's decision is equally crucial, and that techniques such as LIME and Anchors can be attached to operating detectors to help users understand why a piece of media was flagged, effectively turning the detection tool into an educational medium for media literacy. For the system delivery layer, Botnevik et al. (2020) proved through BRENDA that a browser extension architecture can automate the entire credibility assessment process without requiring users to leave the webpage. On the content authenticity front, Tolstykh et al. (2024) introduced GigaCheck, a dual-strategy framework that localizes

AI-generated text spans at the document level, providing a methodology directly applicable to detecting LLM-generated disinformation narratives within social media captions. Addressing the security of the extension layer itself, Hassan et al. (2025) warn that browser extensions with extensive permission sets create significant vectors for permission abuse and cyber threats, an architectural concern that directly informed PRISM's adoption of the privacy-preserving Manifest V3 standard.

In response to these converging challenges, this study will propose PRISM, a multimodal deep learning system for progressive real-time identification of synthetic media and disinformation on social media platforms. The system will integrate a fine-tuned Filipino BERT model, a CNN-ViT hybrid image detector, and a frame-level forensic engine to convert complex visual and textual inputs into transparent, explainable credibility verdicts. By prioritizing browser-based accessibility, the system will bridge the gap between advanced AI forensics and the daily verification needs of the general public, contributing to the achievement of Sustainable Development Goal (SDG) 16: Peace, Justice, and Strong Institutions; SDG 4: Quality Education; and SDG 10: Reduced Inequalities.

Statement of the Problem

The rapid evolution of generative AI has made the creation of high-fidelity synthetic media accessible to the general public, leading to an industrial-scale spread of disinformation. This phenomenon has transformed social media into a major cybersecurity front, with fraud-related losses exceeding USD 1.2 billion globally in 2023 alone (Jha, 2025). Recommendation algorithms compound the problem by disproportionately amplifying misinformation over credible content (Pathak et al., 2023), while current manual fact-checking processes remain insufficient due to their inherent latency. Existing automated forensic tools are often siloed to a single modality and fail

to account for the unique linguistic nuances of Taglish code-switching prevalent in social media discourse.

This study will aim to address the critical lack of an integrated, real-time detection system that can simultaneously verify multiple media types within the Philippine digital landscape. Specifically, this research seeks to answer the following questions:

1. How can a multimodal detection system be architected to simultaneously identify AI-generated text, images, and deepfake videos with high accuracy?
2. To what extent can a Filipino BERT model be fine-tuned to effectively classify "Taglish" disinformation narratives common on local social media?
3. What specific spatial and temporal artifacts are most effective for distinguishing real Filipino political and social media videos from deepfake manipulations?
4. How can a browser-based interface be designed to provide real-time, explainable forensic verdicts without compromising the user's browsing experience or device performance?

Objectives of the Study

The general objective of the study will be to design, develop, and evaluate PRISM, a multimodal deep learning system for progressive real-time identification of synthetic media and disinformation on social media platforms. Specifically, the study will:

Design a multimodal system featuring a Taglish-aware fake news detector, an AI-generated image classifier, and a deepfake video forensic module, all integrated into a unified browser extension and web-based interface;

Develop the PRISM system using Python-based backend frameworks (FastAPI) and deep learning libraries (TensorFlow/PyTorch) for model serving with Supabase serving as the backend platform, while utilizing React for the web-based manual content submission interface and JavaScript (Manifest V3) for the browser extension;

Test the system's functional performance and detection accuracy using a curated dataset of social media disinformation and established media forensic benchmarks;

Evaluate the system's effectiveness based on precision, recall, and F1-score metrics, as well as its user experience through the System Usability Scale (SUS) with social media users; and

Prepare an implementation plan for the pilot deployment of the PRISM browser extension to provide immediate verification tools for the digital community.

Hypothesis

Null Hypothesis (H_0): There will be no significant difference in the detection accuracy and recall of disinformation when using the PRISM multimodal system compared to existing single-modality detection methods. Additionally, the system will not meet the required latency thresholds for real-time browsing on desktop devices.

Alternative Hypothesis (H_1): The PRISM multimodal system will achieve significantly higher detection accuracy and recall in identifying synthetic media and disinformation compared to single-modality baselines. Furthermore, the optimized architecture will allow the system to operate within real-time latency thresholds, providing a seamless and explainable user experience for social media users.

Conceptual Framework of the Study

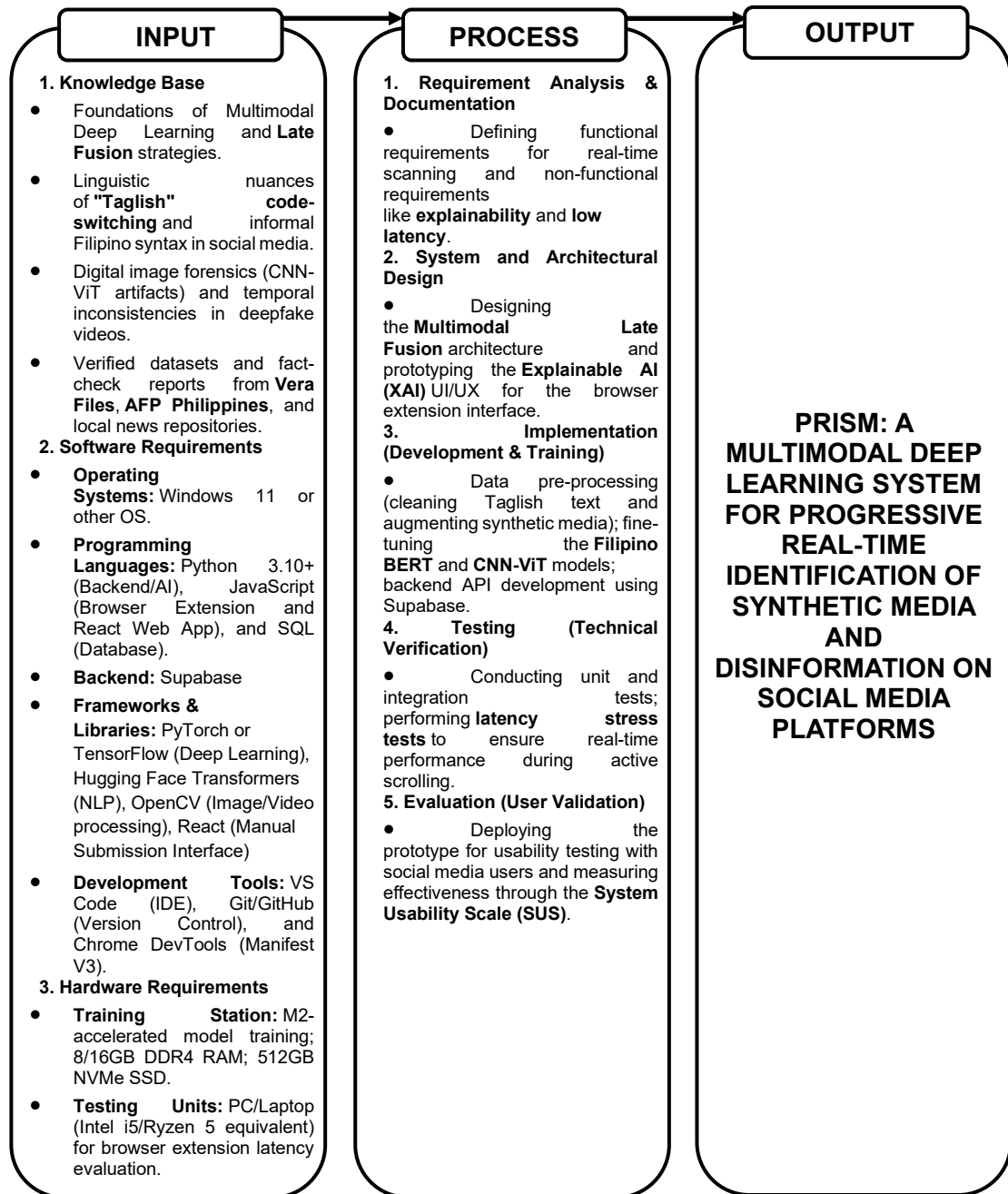


Figure 1. The Conceptual Framework of PRISM

The conceptual framework of this study (see Figure 1) will be anchored in the Input-Process-Output (IPO) model, which will provide a structured approach for the development and evaluation of the system. This model will serve as the

general structure for the study, ensuring that technical and academic requirements are systematically transformed into a functional solution for detecting disinformation.

The Input phase will consist of three critical categories: Knowledge Base, Software Requirements, and Hardware Requirements. The knowledge base includes the theoretical foundations of multimodal deep learning and late fusion strategies necessary for identifying fraudulent activities (Alonso-Bartolome & Segura-Bedmar, 2022). It incorporates the specific linguistic nuances of "Taglish" code-switching to address the lack of language specificity in existing detectors (Diwan et al., 2021). Furthermore, it utilizes digital image forensics for identifying CNN-ViT artifacts and verified fact-check datasets sourced from Vera Files and AFP Philippines for text and image samples, supplemented by established benchmark datasets (e.g., FakeNewsNet) for model pre-training, with preprocessing steps including Taglish normalization, class balancing, and frame-level extraction for video forensics, leveraging the persuasive power of visual elements often found in multimodal fake news (Visweswaran et al., 2024). The software requirements encompass the technical stack, including Supabase for the backend and BERT-based models, which have shown superior performance in text classification (Diwan et al., 2021), React for the web-based manual content submission interface, and JavaScript with Manifest V3 for the browser extension component. The hardware requirements for creating this involves an M2-accelerated training station and consumer-grade desktop units for real-world latency evaluation of both the browser extension and the React web application.

The Process phase will follow a research-oriented software development life cycle tailored for AI integration. It begins with Requirement Analysis and Documentation, focusing on the functional needs for real-time scanning and the non-functional requirements of low latency. The System and Architectural Design stage involves architecting a Late Fusion strategy, where each forensic module, text, image, and video, will be trained independently to its optimum before their confidence scores will be combined at the final decision stage. This approach was selected over early or intermediate fusion because it accommodates posts where only one modality is present (e.g., text-only or image-only content), prevents error propagation between heterogeneous model architectures during training, and allows each module to be debugged and updated independently (Jing et al., 2023). During Implementation, the researchers will conduct data pre-processing and fine-tune the forensic models. This will be followed by Technical Verification through latency stress tests. Finally, the Evaluation stage will utilize the System Usability Scale (SUS) to validate the system's effectiveness, while incorporating Explainable AI (XAI) techniques to ensure the system's decisions are transparent to the end-user (Chuan et al., 2024).

The Output of the study is the developed project, titled PRISM: A Multimodal Deep Learning System for Progressive Real-Time Identification of Synthetic Media and Disinformation on Social Media Platforms. System performance is quantitatively measured using Precision, Recall, and F1-score against single-modality baselines, while user experience will be assessed through the System Usability Scale (SUS), ensuring the system will be validated on both technical and human-centered dimensions. This represents a unified, real-time

forensic tool that will provide transparent credibility verdicts. The final system will be subjected to user evaluation to ensure it successfully bridges the gap between complex AI forensics and the practical verification needs of the Filipino digital community.

Significance of the Study

This study will develop PRISM, a multimodal detection system designed to combat digital deception and the rapid spread of synthetic media on social media platforms. By integrating Taglish-aware NLP, an AI-image classifier, and deepfake forensics into a passive browser extension, the project modernizes how citizens verify content in real time. This will directly address a critical gap in Southeast Asian language specificity and multimodal data coverage identified by Cruz et al. (2020) and Diwan et al. (2021), while responding to the escalating threat of AI-driven electoral disinformation documented by Schipper (2025) and the deepfake-related harms affecting young Filipinos identified by Blancaflor et al. (2023, 2024). The findings will be expected to benefit the following:

Social Media Users and the General Public. The system acts as a "digital shield," providing instant credibility verdicts and explainable signals to help users navigate misinformation and improve long-term media literacy.

Fact-Checking Organizations and Media Practitioners (e.g., Vera Files and AFP Philippines). PRISM will serve as a high-speed preliminary filter, enabling these entities to triage large volumes of flagged content with greater efficiency and consistency.

Government and Educational Agencies. The research provides a structured approach to monitoring digital threats and supports the development of localized media literacy programs tailored to the Philippine context.

Academic and Research Community. This study contributes to the body of knowledge by demonstrating how synergistic fusion models can outperform single-modality tools in accuracy and reliability.

Future Researchers and System Developers. The methodology and architectural framework provide a reference for further studies on multimodal AI fusion and real-time forensic deployment in low-resource settings.

Time and Place of the Study

The study will be conducted from May 2026 to June 2026. The time of the study starts on the day when the proposal will be approved and the researchers begin preparing the necessary instruments for data gathering, up to the day when the researchers are ready to present the final system at the Next Gen Start-Up Competition 2026 at Mapua University - Makati Campus on June 18, 2026. Proposed activities to be implemented in this research include the preparation of the project proposal, system design, prototype development, testing, and evaluation. The digital setting of the study encompasses the social media platforms for which PRISM is designed, namely Facebook, TikTok, and X, ensuring that the features and functionalities of the system properly match the real-world content verification needs of its intended users.

Scope and Limitations of the Study

The scope of this study focuses on the design, development, and evaluation of **PRISM**, a multimodal forensic system specifically engineered to detect AI-generated text, fake news, AI-generated images, and deepfake videos. The investigation covers the analysis of digital content typically found on mainstream social media platforms, namely Facebook, TikTok, and X (formerly Twitter). The research will utilize a **fusion-based architecture** that will integrate three distinct forensic modules: a fine-tuned **Filipino BERT** model for "Taglish" code-switched text detection, a **CNN-Vision Transformer (ViT)** hybrid for image classification, and a frame-level forensic engine for identifying both spatial and temporal inconsistencies in videos. The spatial component analyzes individual frames for pixel-level manipulation artifacts, while the temporal component examines inter-frame patterns including pixel jitter, lip-sync mismatches, and surface frame anomalies that only manifest across time, which are critical for detecting high-quality deepfakes that evade purely static analysis. The study area is primarily digital, utilizing real-world datasets from **Vera Files** and **AFP Philippines** to ensure the system's training will be grounded in actual disinformation narratives circulating within the local landscape.

To measure the system's effectiveness, the researchers will test parameters including **Detection Accuracy, Precision, Recall, and F1-score**, comparing the multimodal results against single-modality baselines. The depth of the investigation includes a comparative analysis of different spatial and temporal artifacts found in deepfake manipulations. For the user evaluation phase, the study will involve a sample of social media users to measure the system's **Explainable AI (XAI)** interface and overall performance. The statistical tools to be employed include the **Confusion Matrix** for evaluating AI classification performance and the **System Usability Scale (SUS)** to quantify user experience and trust levels.

On the other hand, the study is subject to several limitations. Geographically and linguistically, the scope will be restricted to Tagalog and English (Taglish) content; therefore, the system may exhibit reduced accuracy when analyzing other Philippine regional dialects like Bisaya or Ilocano. This limitation will be recognized as increasingly significant, as current trends in Philippine disinformation show a deliberate shift toward regional dialects to evade centralized AI filters. While the prototype is Taglish-centric, the modular architecture of PRISM is explicitly designed for future Dialect Expansion, wherein additional language-specific modules can be trained and integrated without restructuring the core detection pipeline, establishing a scalable pathway toward broader regional language coverage. In terms of sample size and strategy, the training data will be limited to documented cases from existing fact-checking repositories, meaning the system may struggle to identify “Zero-Day” disinformation or synthetic media generated by emerging AI models not yet represented in the training set.

Furthermore, the accuracy and sensitivity of the tools are dependent on the computational power of the user's device and the stability of their internet connection. The "real-time" performance of the **Manifest V3** browser extension may be hindered on low-end consumer hardware or high-latency networks. Availability of facilities will be limited to the researchers' local hardware and cloud-based GPU environments, which may restrict the speed of extensive model epoch training. From an ethical and social perspective, the study will be limited to being a decision-support tool; it does not possess the authority to censor or remove content, and its forensic verdicts will be intended to serve as credibility signals rather than absolute legal evidence.

Definition of Terms

AI-generated Content refers to any digital media, including text or images, that will be produced or significantly altered by artificial intelligence algorithms rather than human creation. In this study, it will serve as the primary target for detection within the multimodal framework to distinguish machine-generated media from human-authored information.

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained deep learning model designed to understand the context of words in a sentence. Operationally, a fine-tuned Filipino version of this model will be utilized as the core natural language processing engine to classify the credibility of social media captions written in Taglish.

Class Activation Map (CAM) is a visualization technique that highlights the specific regions of an image most responsible for a model's classification decision. In this study, CAM-based methods will be integrated into the CNN-ViT image forensic module to generate visual heatmaps that communicate to users which parts of an image were flagged as manipulated or AI-generated.

CNN-ViT Hybrid is a computational architecture that will combine the local artifact-feature extraction capabilities of Convolutional Neural Networks (CNN) with the global semantic relationship modeling of Vision Transformers (ViT). This will describe the specific image forensic model to be developed to identify manipulated visual artifacts in social media photos.

Code-switching is a sociolinguistic phenomenon in which a speaker alternates between two or more languages within a single conversation or utterance. In this study, it refers specifically to the seamless alternation between Tagalog and English that

characterizes social media discourse and defines the linguistic challenge that the Taglish-aware text detection module will be designed to address.

Confusion Matrix is a specific table layout used to visualize and measure the performance of a classification algorithm. In this study, it will be the primary statistical instrument used to calculate the precision, recall, F1-score, and overall accuracy of each forensic detection module.

Deepfake is a general term for synthetic media in which a person in an existing image or video has been replaced with someone else's likeness using deep learning. In this research, it will specifically refer to the manipulated videos that the frame-level forensic engine will seek to identify through spatial and temporal analysis.

Diffusion Model is a class of generative AI architecture that will produce images by learning to reverse a gradual noising process. In this study, diffusion model outputs will represent a primary generative paradigm targeted by PRISM's image classifier alongside GAN-generated images.

Disinformation refers to false information that will be deliberately created and spread to influence public opinion. Operationally, it will refer to the misleading narratives and fabricated media circulating on social media platforms that the PRISM system will be designed to flag.

DistilBERT-Tagalog is a compressed variant of the Filipino BERT model produced through Knowledge Distillation, retaining over 95% of classification performance at approximately 40% of the original model size. In this study, it will be the specific model deployed within the PRISM system to enable real-time, lightweight server-assisted text inference.

Explainable AI (XAI) is a set of processes and methods that will allow human users to comprehend and trust the results created by machine learning algorithms. In this study, XAI will be implemented through visual heatmaps and text highlighting (via LIME and Anchors) to show users the specific rationale behind each forensic verdict.

F1-Score is a statistical measure that will combine both precision and recall into a single value. In this research, it will serve as the definitive metric used to prove that the multimodal PRISM system is more reliable than existing single-modality tools.

Fake News refers to fabricated or deliberately misleading information presented in the format of legitimate news content and distributed through media channels. In this study, it will be used interchangeably with disinformation to describe the class of false or manipulative content that PRISM's text and multimodal detection modules will be trained to identify on social media platforms.

Fact-checking refers to the process of verifying the accuracy of claims against credible, primary sources. In this study, it will describe the editorial process performed by organizations such as Vera Files and AFP Philippines, whose verified outputs will serve as the ground-truth labels for training and evaluating PRISM's forensic modules.

FastAPI is a high-performance web framework for building APIs with Python. In this study, it will serve as the dedicated AI inference layer, receiving media inputs from the client interfaces and executing the compute-intensive BERT, CNN-ViT, and deepfake forensic models to return credibility verdicts.

Frame-level Forensic Engine is the video analysis component of PRISM that will detect deepfake manipulations by examining both individual video frames for spatial pixel-level artifacts and the sequential relationships between frames for temporal inconsistencies.

Generative Adversarial Network (GAN) is a class of deep learning architecture consisting of a generator and a discriminator. In this study, GAN-generated images will constitute a primary category of synthetic visual content that the image forensic module will be trained to detect.

Inference Latency refers to the time it will take for an AI model to process an input and produce an output. Operationally, this is the processing delay the study will aim to minimize to ensure a seamless, real-time user experience during active browsing.

Knowledge Distillation is a model compression technique in which a smaller “student” model will be trained to replicate the behavior of a larger “teacher” model, enabling lightweight deployment with reduced computational requirements.

Late Fusion is a multimodal integration strategy where individual forensic models will be trained separately, and their classification confidence scores will be combined only at the final decision stage to produce a unified credibility verdict.

LIME (Local Interpretable Model-Agnostic Explanations) and Anchors are Explainable AI techniques that will be applied to PRISM’s text detection module to highlight specific words or phrases that most strongly influenced a forensic verdict.

Manifest V3 is the current generation of the Chrome extension platform architecture. It will serve as the technical foundation for the PRISM browser extension, dictating how the system will interact with web content while maintaining strict user privacy and security standards.

Multimodal Detection refers to the simultaneous analysis of multiple types of data, such as text, images, and video. Operationally, this will refer to the system’s

capability to evaluate a social media post's caption and its embedded media concurrently to increase detection accuracy.

Prototyping Model is a System Development Life Cycle methodology in which a functional prototype will be constructed rapidly, evaluated, and iteratively refined through a loop of construction and testing until all requirements will be met.

Real-time Identification is the immediate processing and classification of data as it will be received by the system. In this study, it will describe the system's ability to provide forensic verdicts as a user scroll through their social media feed without significant lag.

Spatial and Temporal Artifacts refer to visual inconsistencies serving as forensic evidence. Spatial artifacts are pixel-level anomalies found within individual frames; temporal artifacts are inter-frame inconsistencies, such as pixel jitter, that will only manifest across video frame sequences.

Supabase is an open-source Backend-as-a-Service (BaaS) platform built on PostgreSQL. In this study, Supabase will provide the PRISM system with structured data storage, user authentication, real-time subscriptions, and API routing through Edge Functions.

Synthetic Media is a broad term for any media that will have been fully or partially generated or manipulated by artificial intelligence. In this study, it will serve as the general category of content targeting forensic verification, encompassing deepfakes and AI-generated text.

System Usability Scale (SUS) is a standardized ten-item tool used to measure usability. In this study, it will be the questionnaire used to quantify user experience and trust levels among social media users.

Taglish is a linguistic phenomenon in the Philippines characterized by code-switching between Tagalog and English. Operationally, it will define the linguistic scope and the primary dataset used to train the system's natural language processing module.

Transfer Learning is a machine learning technique in which a model pre-trained on a large dataset will be fine-tuned on a smaller, domain-specific dataset. This will be the foundational training methodology for all of PRISM's forensic modules.

Vision Transformer (ViT) is a model architecture that applies transformer mechanisms to image processing. In this study, it will be used to capture intricate global visual dependencies within social media images to detect sophisticated AI-driven manipulations.

REVIEW OF RELATED LITERATURE

Local Literature

Disinformation and Fake News in the Philippines

Ong and Cabañes (2018) conducted an investigative study examining the organizational structure and operational mechanics of disinformation networks in the Philippines. Through deep narrative interviews with workers positioned at multiple levels of fake news production hierarchies, the study revealed that disinformation in the Philippines is not spontaneous but systematically produced through coordinated labor arrangements involving social media managers, content creators, and community organizers working on behalf of political clients. The study documented the specific persuasive techniques that architects of networked disinformation deploy to mobilize populist sentiment, including the deliberate exploitation of emotional triggers, platform algorithms, and existing social divisions. The findings establish that Filipino disinformation is structurally distinct from its counterparts in other countries due to the depth of its integration into political and commercial ecosystems. For the present study, Ong and Cabañes (2018) provide the foundational sociopolitical context that motivates the development of PRISM: the threat being addressed is not isolated content but an organized, platform-native disinformation infrastructure that requires an equally systematic automated response.

Valdez, Ugalingan, and Garinto (2021) examined the challenge of combating misinformation in an online English language classroom setting in the Philippines during the COVID-19 pandemic. The study documented how the simultaneous disruption of education and the surge of health-related misinformation created a compounded challenge for Filipino teachers and learners, who were forced to navigate unreliable information environments while adjusting to fully online instruction. The authors argued that incorporating multimodal resources, structuring arguments, and

identifying opportunities for critical engagement are essential components of effective misinformation resistance strategies. A central finding of the study is that Filipino internet users, particularly students, lack structured frameworks for evaluating the credibility of digital content, making them vulnerable to sophisticated disinformation even when they are aware of its general existence. This finding directly informs the design rationale of PRISM's Explainable AI interface: a detection system that simply flags content without explaining its reasoning does not build the media literacy that Filipino users need for long-term resilience against disinformation.

Schipper (2025) examined the dual role of artificial intelligence in shaping electoral integrity and combating misinformation in the context of the 2025 Philippine elections. Drawing on comparative case studies from the Philippines, Taiwan, and India, the commentary documented how generative AI tools were actively weaponized to produce revisionist narratives, synthetic media, and algorithmically amplified disinformation targeting Filipino voters. The study further evaluated the effectiveness of countermeasures deployed across these three countries, finding that real-time AI fact-checking tools, public misinformation tiplines, and transparency mandates for AI-generated content were among the most effective interventions. Critically, the study concluded that no single technological solution is sufficient and that scalable, adaptable AI detection tools must be integrated within broader regulatory frameworks and digital literacy initiatives. For the present study, Schipper (2025) provides the most current and contextually immediate evidence that the threat environment PRISM will be designed to address is active, escalating, and directly affecting democratic processes in the Philippines.

Deepfakes and Synthetic Media in the Philippine Context

Blancaflor, Anave, Cunanan, Frias, and Velarde (2023) conducted a literature review and survey study examining Filipino awareness of deepfakes and the state of deepfake legislation and regulation in the Philippines. Using a survey instrument administered to 165 respondents, the study found a statistically significant correlation between social media exposure and both awareness of deepfakes and the ability to identify them. However, the study also revealed that the majority of respondents were unaware of existing legislation governing deepfakes, and that those with prior exposure to deepfake content were most likely to recognize the inadequacy of current regulatory frameworks. The findings indicate that while awareness is growing, it will be driven by organic exposure to harmful content rather than by structured educational or regulatory initiatives. This gap between growing exposure and inadequate institutional response is precisely the environment in which PRISM operates: a system that provides real-time credibility signals at the point of content consumption addresses the moment of vulnerability that neither legislation nor media literacy programs can currently reach in time.

Blancaflor, Garcia, Magno, and Vilar (2024) conducted a meta-analysis investigating the use of deepfake technology in blackmailing and revenge pornography cases targeting Filipino internet users. The study identified the structural factors that make deepfakes particularly effective as tools of cyber sextortion, including the photorealistic quality of modern face-swap technologies, the psychological pressure generated by the threat of non-consensual intimate image distribution, and the asymmetric power dynamics between perpetrators who understand the technology and victims who do not. The study further evaluated existing regulatory structures and proposed detection and blocking measures for pornographic deepfakes, emphasizing that technical countermeasures must be accessible to ordinary users rather than requiring specialized expertise. The findings of Blancaflor et al. (2024) extend the harm

profile of deepfakes in the Philippine context beyond political disinformation into direct personal harm, reinforcing the urgency of developing user-accessible deepfake detection tools that do not require users to possess technical knowledge to benefit from them.

Filipino NLP and Low-Resource Language Detection

Cruz, Tan, and Cheng (2020) addressed the fundamental resource scarcity problem in Filipino fake news detection by constructing the first expertly curated benchmark dataset for fake news classification in Filipino, which they named Fake News Filipino. The study benchmarked multiple transfer learning approaches on this dataset and demonstrated that multitask transfer learning, augmented with auxiliary language modeling losses, achieves accuracy of up to 96 percent on Filipino fake news classification while requiring only small amounts of labeled training data. The results demonstrated that transformer-based transfer learning generalizes effectively across different types of Filipino news content, including political news, entertainment news, and opinion articles, without requiring separate models for each category. For the present study, Cruz et al. (2020) provides both a direct methodological precedent and a dataset reference for the Filipino BERT module within PRISM. The finding that multitask transfer learning can overcome data scarcity in the Filipino context is the foundational justification for the fine-tuning approach adopted in this study rather than training a detection model from scratch.

Diwan, Mittal, Khare, Bharadwaj, and Dhamecha (2021) examined the challenge of developing NLP and automatic speech recognition systems for low-resource languages and code-switching environments, arguing that most advances in these fields will be powered by data availability and therefore benefit only a small fraction of the world's languages. The study described four research projects

addressing low-resource and code-switching challenges, including transliteration-based transfer learning, multilingual ASR challenge organization, speech-grounded transliteration, and linguistically inspired vocabulary reduction. A central argument of the study is that developing less data-intensive techniques for low-resource and code-switching languages demands radically new approaches that go beyond simply scaling existing high-resource methods. This argument is directly relevant to the Taglish module of PRISM: social media content will be characterized by seamless code-switching between Tagalog and English within single sentences, a phenomenon that standard English-trained NLP models are not designed to handle. The strategies described by Diwan et al. (2021) for bridging high-resource and low-resource language capabilities through transfer learning inform the architectural decisions in the present study's text detection module.

Foreign Literature

Multimodal Fake News Detection

Visweswaran, Mohan, Kumar, and Soman (2024) proposed a fusion-based methodology for multimodal fake news identification that integrates Text Graph Convolutional Neural Networks (TextGCN) and Vision Transformers (ViT) to simultaneously process text and image modalities. The TextGCN component will capture intricate structural dependencies among words and phrases by representing text as a graph, while the ViT component extracts visual features from associated images. A fusion mechanism then integrates these modalities to yield superior embeddings compared to either modality alone. Experimental results demonstrated an accuracy of 94.17 percent using a neural network after fusion, substantially outperforming single-modality baselines. The study established that unlike traditional news articles that predominantly rely on textual content, multimodal fake news

leverages the persuasive power of visual elements, making unimodal detection approaches fundamentally insufficient. For the present study, Visweswaran et al. (2024) provides the architectural justification for PRISM's multimodal fusion design and directly motivates the integration of both text and image forensic modules into a unified detection pipeline.

Jing, Wu, Sun, Fang, and Zhang (2023) proposed a progressive fusion network (MPFN) for multimodal disinformation detection that addresses the limitation of existing methods that exploit only deep features of multimodal information, leading to significant loss of valid information at shallow feature levels. The MPFN captures representational information from each modality at multiple levels and achieves fusion between modalities at both the same level and across different levels using a mixer architecture. A Swin Transformer structure will be used as the visual feature extractor, and image frequency domain information will be incorporated at different levels for fine-grained modeling. Experimental results on the Weibo and Twitter datasets showed that MPFN achieved 83.3 percent accuracy on the Twitter dataset, representing at least a 4.3 percent improvement over prior state-of-the-art methods. For the present study, Jing et al. (2023) provides the theoretical basis for the late fusion strategy adopted in PRISM and demonstrates that capturing information at multiple feature levels is critical for achieving high classification precision across diverse real-world content types.

Alonso-Bartolome and Segura-Bedmar (2022) performed a comprehensive comparison of unimodal and multimodal approaches for fake news classification on the Fakeddit dataset, which contains over one million Reddit posts classified into six categories of fake news. The study evaluated CNN, BiLSTM, and BERT architectures for text-only classification, and proposed a multimodal CNN architecture that combines text and image data. Results showed that the multimodal CNN approach achieved the best performance with an accuracy of 87 percent, compared to 78 percent for BERT using text alone, demonstrating that exploiting both text and image data significantly

improves fake news detection performance. The study further found that categories such as Manipulated Content, Satire, and False Connection benefit particularly strongly from the inclusion of image data, while text-only approaches fail to adequately distinguish these categories. For the present study, Alonso-Bartolome and Segura-Bedmar (2022) provides empirical evidence that multimodal approaches consistently outperform text-only detection across diverse fake news categories, directly justifying the architectural choice to integrate image and text forensic modules in PRISM.

Cui, Wang, and Lee (2019) investigated the role of user comments and their latent sentiments in detecting fake news, developing the Sentiment-Aware Multi-Modal Embedding (SAME) framework. The SAME framework uses multi-modal networks to handle heterogeneous data modalities, an adversarial mechanism to learn semantically meaningful embedding spaces per data source, and a novel regularization loss to bring embeddings of relevant content pairs closer together. Validation on the PolitiFact and GossipCop datasets demonstrated that SAME significantly outperforms state-of-the-art methods by incorporating social context signals that purely content-based detectors ignore. The study established that user comments carry latent sentiment information that distinguishes fake content from reliable content in ways that the content itself does not always reveal. For the present study, Cui et al. (2019) provides the theoretical grounding for considering social context as a complementary signal in multimodal detection systems, informing future extension directions for PRISM beyond its core text, image, and video modules.

Szczepanski, Pawlicki, Kozik, and Choras (2021) proposed a novel explainability approach for BERT-based fake news detectors that can be attached to existing operating detectors without requiring extensive architectural modifications. The approach applies two Explainable AI techniques, Local Interpretable Model-Agnostic Explanations (LIME) and Anchors, to BERT-based models trained on fake news data consisting of short text fragments such as tweets and news headlines. The

study demonstrated that these surrogate-type explanation methods produce locally faithful explanations of model predictions, allowing non-specialist users to understand why a given piece of content was classified as fake or real. A central argument of the study is that high model performance is no longer sufficient for real-life deployment scenarios and that the explainability of the system's decision is equally crucial. For the present study, Szczepanski et al. (2021) provides the methodological foundation for the Explainable AI interface in PRISM, directly informing the implementation of visual heatmaps and text highlighting as mechanisms for communicating detection rationale to social media users.

Jha (2025) proposed a novel multimodal detection framework integrating Natural Language Processing, Computer Vision, and Graph Neural Networks to identify fraudulent activities across text, image, and relational network dimensions on social media platforms. The study reported that social media fraud caused over USD 1.2 billion in reported losses in 2023 alone, and that existing rule-based detection methods fail to address the dynamic and multimodal nature of these threats. Experiments conducted on real-world datasets including the Twitter Bot Dataset, Deepfake Detection Challenge, and PhishTank demonstrated an 18 percent improvement in detection accuracy compared to traditional systems. For the present study, Jha (2025) provides both the quantitative scale of the threat environment and a working example of a multimodal detection architecture that simultaneously addresses text, image, and network-based fraud signals, directly paralleling PRISM's multi-module design.

Deepfake Video Detection

Heidari, Navimipour, Dag, and Unal (2023) conducted a systematic and comprehensive review of deepfake detection methods using deep learning algorithms,

categorizing detection approaches based on their application domains: video detection, image detection, audio detection, and hybrid multimedia detection. The review established that Convolutional Neural Networks are the most frequently employed deep learning method in published deepfake detection studies, and that the majority of existing research focuses on video deepfake detection as the primary application domain. The study further identified that most publications focus on optimizing a single performance parameter, with detection accuracy receiving the most attention, while computational efficiency and generalization to unseen manipulation methods remain significantly under addressed. Key weaknesses identified across existing systems include overconfidence in detection outputs, poor generalization to new generation architectures, and limited performance on real-world content that differs from controlled dataset conditions. For the present study, Heidari et al. (2023) provides the comprehensive methodological landscape against which PRISM's frame-level forensic engine will be positioned, and their identification of real-time deployment as an open research problem directly motivates the latency optimization component of the present study.

Akhtar and colleagues (2024) published a survey paper analyzing the challenges and opportunities in deepfake video detection, classifying the difficulties encountered by detection systems into three major categories: data challenges, including unbalanced datasets and insufficient labeled training data; training challenges, including the high computational resources required for model development; and reliability challenges, including overconfidence in detection methods and the rapid emergence of new manipulation techniques that outpace existing detectors. The survey further critically evaluated existing deepfake datasets and emphasized that the quality and diversity of training data are the primary determinants of real-world detection performance. The study identified developing robust models for real-time detection as the most significant open research gap in the field. For the

present study, Akhtar et al. (2024) provides the most current and comprehensive mapping of the deepfake detection challenge landscape, directly informing the dataset curation strategy and the real-time performance requirements established in PRISM's system design.

AI-Generated Image Detection

Park, Na, and Choi (2024) conducted a performance comparison and visualization study of AI-generated image detection methods, classifying existing detection approaches by the image features they exploit and evaluating their generalization performance across images generated by different types of generative models. The study selected five detection methods for evaluation and used a vision transformer as an additional baseline, testing performance across GAN-generated, diffusion-generated, and transformer-generated image categories. The key finding was that artifact-feature-based detection methods perform well on GAN and real images, while image-encoder-feature-based detection methods perform better on diffusion model and transformer-generated images, demonstrating that no single detection approach generalizes equally well across all generative paradigms. For the present study, Park et al. (2024) directly informs the architecture of PRISM's CNN-ViT hybrid image classifier by demonstrating that robust AI-generated image detection requires combining detection approaches that address both artifact-level features and encoder-level semantic features.

Baraheem and Nguyen (2023) proposed a CNN-based framework for detecting AI-generated images produced by Generative Adversarial Networks, using transfer learning applied to a diverse collection of GAN-generated images across different tasks and architectures. The study integrated several Class Activation Map techniques to identify the discriminative image regions that guided the classification model's

decisions, providing both detection capability and visual explainability. Their best detector, a pre-trained EfficientNetB4 fine-tuned on the Real or Synthetic Images dataset, achieved 100 percent accuracy on the in-distribution dataset and superior performance on external datasets. For the present study, Baraheem and Nguyen (2023) provides the transfer learning methodology and dataset construction principles applied in PRISM's image forensic module, and their integration of Class Activation Maps directly parallels the visual heatmap explainability mechanism implemented in the present study's user interface.

Real-Time Detection Systems

Zhang, Gupta, Qin, and Zhou (2023) proposed a computational approach to detecting fake news in a real-time manner, leveraging event and topic extraction techniques coupled with a topic merging mechanism to process streaming news data efficiently. The approach incorporates a two-stage procedure to optimize the cold-start ratio between initial data batches and subsequent ones, improving memory management during the processing of streaming data. Computational experiments demonstrated that the proposed approach reduces the number of topics by 19.76 percent and the number of data clusters by 26.92 percent compared to baseline methods while achieving faster detection times. For the present study, Zhang et al. (2023) provides the computational design principles for handling streaming social media data in real time, directly informing the backend processing architecture of PRISM and the latency optimization strategies implemented to ensure the system remains responsive during active user browsing sessions.

Tian, Zhang, and Peng (2020) designed and developed FakeFinder, a mobile application for detecting fake news from the live Twitter stream using a device-based prediction model built on the compact language model ALBERT. The study

demonstrated that a small language model can achieve real-time, accurate fake news detection directly on a mobile device without requiring server-side inference, addressing the latency and connectivity constraints that prevent cloud-dependent systems from providing seamless real-time performance. The authors established that the prediction model must be small enough to allow fast inference on mobile hardware while remaining accurate enough to be practically useful, identifying this size-accuracy tradeoff as the central design challenge for mobile deployment. For the present study, Tian et al. (2020) provides both the architectural template for PRISM's implementation on its website extension and the empirical validation that on-device or lightweight server-assisted inference is a viable strategy for achieving real-time fake news detection within the resource constraints of consumer.

Browser Extension Approaches

Botnevik, Sakariassen, and Setty (2020) proposed BRENDA, a browser extension designed to automate the entire process of credibility assessment for false claims encountered during web browsing. BRENDA uses a deep neural network architecture to automatically identify fact-check-worthy claims within web page content and classifies them with credibility assessments that will be presented to the user alongside supporting evidence without requiring the user to leave the webpage. The system demonstrated that a browser extension architecture can provide seamless, passive fact-checking that integrates into the existing browsing workflow rather than requiring users to actively initiate separate verification processes. For the present study, Botnevik et al. (2020) provides the primary architectural reference for PRISM's browser extension component, establishing the precedent that deep neural network-powered credibility assessment can be delivered passively within the browser environment using a Manifest V3-compatible extension design.

Hassan, Zeeshan, and Ahmed (2025) analyzed the security vulnerabilities introduced by browser extensions operating with extensive permission sets, documenting how third-party applications that require broad access to browser functionality create significant vectors for permission abuse, data exposure, and exploitation by malicious actors. The study identified that the convenience provided by browser extensions often comes at a considerable security cost, as the permissions required for functionality can be misused to expose users to a range of cyber threats including data harvesting and content injection. For the present study, Hassan et al. (2025) provides the security design rationale for PRISM's adoption of the Manifest V3 extension standard, which imposes stricter permission controls and reduces the attack surface available to the extension compared to earlier extension architectures, ensuring that the detection system does not itself introduce the vulnerabilities it will be designed to help users navigate.

LLM-Generated Content Detection

Tolstykh, Tsybina, Yakubson, Gordeev, Dokholyan, and Kuprashevich (2024) introduced GigaCheck, a dual-strategy framework for AI-generated text detection that operates at both the document level and the span level. At the document level, the framework leverages representation learning of fine-tuned large language models to distinguish AI-authored from human-authored text with high data efficiency. At the span level, a novel structural adaptation treats generated text segments as objects, integrating a DETR-like vision model with linguistic encoders to achieve precise localization of AI-generated intervals within mixed-authorship documents. Experimental results across three classification and three localization benchmarks confirmed the robustness of the approach across diverse generation styles and source

models. For the present study, Tolstykh et al. (2024) provides the theoretical and methodological basis for PRISM's detection of LLM-generated disinformation text within social media captions, where AI-generated content may be mixed with human-authored context in ways that require span-level rather than document-level detection precision.

Social Media Algorithms and Misinformation Spread

Pathak, Spezzano, and Pera (2023) conducted an in-depth analysis of the relationship between recommender algorithms and misinformation propagation on social networks, using the PolitiFact FakeNewsNet and HealthStory FakeHealth datasets to examine how well-known recommendation algorithms will be impacted by and contribute to misinformation spread on Twitter. The study found that popularity-based and network-based recommender algorithms contribute the most to misinformation diffusion, and that users identified as superspreaders directly amplify the impact of algorithmic recommendations on misinformation spread. A simulation component further demonstrated that misinformation diffusion patterns triggered by algorithmic suggestions are measurably distinct from organic diffusion, confirming that platform recommendation systems are an independent amplification mechanism for false content beyond what user behavior alone would produce. For the present study, Pathak et al. (2023) establishes the platform-level context within which PRISM must operate: the system is not simply detecting isolated pieces of false content but is intervening against content that is actively being amplified by the same recommendation systems Filipino users interact with during every browsing session, making real-time passive detection at the point of content display the most effective deployment strategy.

RELATED STUDIES

Local Studies

Deepfake Awareness and Regulation in the Philippines

Blancaflor et al. (2023) surveyed 165 Filipino respondents to measure awareness of deepfakes and knowledge of existing legislation. Their findings revealed that Filipino exposure to deepfakes through social media correlates with the ability to recognize them, but that most respondents could not identify specific laws governing deepfake creation or distribution. The study concluded that legislative frameworks alone are insufficient without accompanying technical tools and public awareness programs. This finding positions PRISM as a practical complement to legislative efforts: while laws address accountability after harm has occurred, a real-time detection system addresses the moment of exposure before harm propagates.

Blancaflor et al. (2024) investigated deepfake-enabled blackmail and revenge pornography in the Philippines through a meta-analysis of documented cases. The study mapped the tactics used by perpetrators, the psychological mechanisms through which victims will be coerced, and the existing legal remedies available in the Philippine context. It concluded that effective countermeasures must include both regulatory structures and technical detection tools accessible to ordinary users. For the present study, Blancaflor et al. (2024) is the most direct local evidence that the deepfake harm profile in the Philippines extends beyond political disinformation into direct personal victimization, establishing a broader user safety justification for the deepfake detection module in PRISM.

Filipino Fake News Dataset and Classification

Cruz et al. (2020) developed the Fake News Filipino dataset and demonstrated that multitask transfer learning achieves 96 percent accuracy on Filipino fake news classification using only small amounts of labeled data. As a study, rather than just a literature reference, Cruz et al. (2020) provides the most direct precedent for the Taglish text classification module in PRISM: it demonstrates that a Filipino fake news classifier is technically achievable with existing tools, establishes the benchmark accuracy that the present study aims to approach or exceed under real-world social media conditions, and provides the dataset against which PRISM's text module can be evaluated.

Misinformation Awareness in Filipino Educational Settings

Valdez et al. (2021) documented their classroom-based experience teaching misinformation resistance strategies to Filipino students in an online learning environment during the COVID-19 pandemic. As a study of applied practice rather than a purely theoretical contribution, it provides evidence that Filipino users benefit from structured guidance in evaluating digital content credibility and that multimodal instructional approaches are more effective than text-only frameworks. For the present study, Valdez et al. (2021) reinforces the design decision to include Explainable AI verdicts in PRISM's interface rather than simply presenting a binary true or false classification, as the study demonstrates that understanding the basis for a credibility judgment is a prerequisite for developing lasting media literacy skills among Filipino users.

Foreign Studies

Multimodal Detection System Architectures

Visweswaran et al. (2024) demonstrated through their TextGCN-ViT fusion system that integrating graph-based text representations with visual transformer features produces detection accuracy substantially exceeding single-modality baselines, with results validated on real-world multimodal fake news datasets. As a study rather than a purely theoretical contribution, Visweswaran et al. (2024) provides the closest architectural parallel to PRISM's multimodal fusion design. The present study extends this line of research by incorporating a video forensic module alongside text and image components, and by additionally targeting the Taglish linguistic environment prevalent in social media.

Jing et al. (2023) implemented and evaluated the MPFN on the Weibo and Twitter datasets, achieving state-of-the-art performance through progressive multi-level feature fusion. As an implemented and evaluated system, MPFN provides a direct performance benchmark against which PRISM's multimodal detection results can be compared, and its demonstration that fine-grained modeling through level-wise fusion outperforms coarser fusion approaches informs the architectural specification of the late fusion strategy in the present study.

Browser Extension Implementation

Botnevik et al. (2020) implemented and evaluated BRENDA as a functioning browser extension for automated fake news detection. The demonstration confirmed that a browser extension can successfully integrate deep neural network-powered credibility assessment into the standard web browsing workflow without requiring meaningful changes to user behavior. For the present study, BRENDA is the closest existing system-level precedent for PRISM's browser extension component. The

primary distinction is that BRENDA addresses text-based claim verification while PRISM extends the browser extension paradigm to simultaneous multimodal detection of text, images, and video content within social media feeds.

Real-Time Lightweight Detection Systems

Tian et al. (2020) implemented FakeFinder as a functioning mobile application and validated its real-time performance on live Twitter data. The system demonstrated that compact language model architectures can deliver accurate fake news detection within the computational constraints of mobile devices, providing empirical validation for the on-device or lightweight inference approach adopted in PRISM's website extension. The present study extends the FakeFinder approach by incorporating image and video detection capabilities alongside text classification within a single extension, by providing a React-based web interface for manual content submission, and by targeting Taglish content rather than English-only Twitter posts.

Real-Time Detection Performance

Zhang et al. (2023) implemented and benchmarked a real-time fake news detection system on streaming data, demonstrating measurable improvements in processing speed and topic reduction efficiency compared to existing approaches. Their computational experiments provide the performance benchmarking methodology and baseline metrics against which PRISM's real-time processing performance can be evaluated during the latency stress testing phase of the present study.

Deepfake Detection Methods

Heidari et al. (2023) and Akhtar et al. (2024) collectively reviewed the existing landscape of deepfake detection implementations, cataloging the datasets, architectures, and evaluation protocols used across published systems. These reviews serve as the methodological reference for the frame-level forensic engine in PRISM, establishing the standard spatial and temporal artifact analysis approaches against which the present study's deepfake detection module will be designed and evaluated.

AI-Generated Image Detection Performance

Park et al. (2024) and Baraheem and Nguyen (2023) collectively provide the implementation precedents for AI-generated image detection in PRISM. Park et al. (2024) establishes the comparative performance profile of different detection method categories, while Baraheem and Nguyen (2023) provides the specific transfer learning approach and Class Activation Map visualization methodology implemented in the present study's CNN-ViT image classifier.

LLM-Generated Text Detection

Tolstykh et al. (2024) implemented and validated GigaCheck across six benchmarks, demonstrating robust performance in both document-level classification and span-level localization of AI-generated text. The span-level localization capability demonstrated in GigaCheck is the most direct precedent for PRISM's ability to identify AI-generated segments within social media captions that may contain both machine-generated disinformation and human-authored context, providing the methodological template for the text forensic module's granular analysis capability.

Social Media Algorithm Amplification

Pathak et al. (2023) conducted simulation-based experiments to quantify how different recommendation algorithms amplify misinformation diffusion beyond what organic user behavior produces. The simulation results provide empirical quantification of the urgency driving PRISM's real-time passive detection design: because misinformation will be amplified algorithmically at speeds that manual fact-checking cannot match, automated detection at the point of content display is the only technically viable intervention strategy for the social media environment.

RESEARCH GAP

The foregoing review of related literature and studies reveals that while substantial technical progress has been made in multimodal fake news detection, deepfake forensics, AI-generated image classification, and browser-based detection deployment, critical gaps remain that the existing body of work has not addressed.

In the local context, existing studies on Philippine disinformation document the organizational structure of fake news networks, the legislative inadequacy of current deepfake regulation, and the media literacy deficits of Filipino internet users. However, none of the reviewed local studies has produced a functional computational system capable of automatically detecting disinformation in the Taglish linguistic environment that characterizes social media. Cruz et al. (2020) established a benchmark dataset and demonstrated the feasibility of Filipino fake news text classification, but did not extend this work to image or video forensics. Blancaflor et al. (2023, 2024) documented the deepfake harm profile in the Philippines but did not propose or implement a technical detection solution. No existing local study has developed a system that integrates text, image, and video forensic modules into a unified, real-time, user-accessible tool designed for the social media environment.

In the foreign literature, the multimodal detection systems reviewed by Visweswaran et al. (2024), Jing et al. (2023), and Alonso-Bartolome and Segura-Bedmar (2022) achieve strong results on English-language datasets but do not address the code-switching linguistic patterns that characterize Taglish content. The deepfake detection systems reviewed by Heidari et al. (2023) and Akhtar et al. (2024) will be evaluated on controlled benchmark datasets that do not represent the diversity of manipulation techniques found in Filipino political and social media videos. AI-generated image detection methods reviewed by Park et al. (2024) and Baraheem and Nguyen (2023) focus on GAN and diffusion model outputs in isolation and have not been integrated with text and video detection within a single real-time system. BRENDA (Botnevik et al., 2020) demonstrated browser extension-based fake news detection but addressed only text-based claim verification and does not handle image or video content. FakeFinder (Tian et al., 2020) demonstrated mobile fake news detection but will be limited to text classification on English-language Twitter content.

No existing study, local or foreign, has developed a system that simultaneously addresses Taglish fake news text detection, AI-generated image classification, and deepfake video forensics within a unified architecture delivered through a browser extension with an Explainable AI interface designed for non-technical social media users. The present study directly addresses this convergence of unmet needs by proposing PRISM as the first integrated, real-time, multimodal disinformation and synthetic media detection system specifically designed for the social media context broadly, with particular attention to the Filipino digital landscape.

SYNTHESIS

The literature and studies reviewed in this chapter collectively establish the empirical foundation, technical justification, and contextual necessity for the development of PRISM. The local literature confirms that the Philippines is a high-priority environment for disinformation detection: Ong and Cabañes (2018) established that disinformation is systematically produced within organized networks; Schipper (2025) documented that AI-generated synthetic media actively threatened the integrity of the 2025 Philippine elections; and Blancaflor et al. (2023, 2024) demonstrated that deepfakes are causing direct personal harm to Filipino users in the absence of accessible technical countermeasures. Valdez et al. (2021) further established that Filipino users require not only detection but explanation, as media literacy development depends on understanding the basis for credibility judgments rather than simply receiving verdicts.

The foreign literature establishes the technical foundations for each of PRISM's core modules. Visweswaran et al. (2024) and Jing et al. (2023) validate multimodal fusion as the superior detection architecture for content that combines text and visual elements. Szczepanski et al. (2021) establishes explainability as a non-negotiable requirement for real-world deployment of AI-based detection systems. Heidari et al. (2023) and Akhtar et al. (2024) map the deepfake detection landscape and identify real-time performance as the primary open challenge. Park et al. (2024) and Baraheem and Nguyen (2023) demonstrate that CNN-based transfer learning with visualization techniques provides both accuracy and interpretability for AI-generated image detection. Tolstykh et al. (2024) provides the span-level text forensic methodology for detecting LLM-generated content within social media captions. Pathak et al. (2023) establishes that platform recommendation algorithms actively amplify misinformation, making real-time passive detection the only viable intervention strategy at scale.

The related studies provide the implementation precedents and performance benchmarks for the present study. Cruz et al. (2020) established the dataset and benchmark accuracy for Filipino fake news text classification. Botnevik et al. (2020) demonstrated browser extension delivery of deep neural network-powered credibility assessment. Tian et al. (2020) demonstrated real-time fake news detection using compact, lightweight model architectures suited for low-latency deployment. Zhang et al. (2023) demonstrated real-time processing of streaming fake news data at computational scales relevant to active social media browsing.

Taken together, these works establish a clear progression from the contextual (why disinformation detection is urgent for social media users) to the technical (what architectures and methods are validated for multimodal detection) to the applied (how similar systems have been built and what performance they achieve). This progression leads directly to PRISM's design: a late fusion multimodal detection system integrating a fine-tuned Filipino BERT module, a CNN-ViT hybrid image classifier, and a frame-level video forensic engine, delivered through a Manifest V3 browser extension with an Explainable AI verdict interface. PRISM represents the convergence point of this body of work, filling the gap that separates the rich global literature on multimodal detection from a functional, accessible, Taglish-aware system designed for the specific disinformation threats facing social media users today.

METHODOLOGY

Research Design

This study will adopt a developmental research design for the purpose of designing, developing, and evaluating PRISM, a real-time multimodal detection system for disinformation and synthetic media on social media platforms. Developmental research, as described in the literature on educational and technological inquiry, will be concerned with the systematic process of designing, creating, and testing functional products or processes that address a clearly defined problem (Richey & Klein, 2007). This design is the most appropriate for the present study because the primary output is a functioning software system rather than a purely theoretical or descriptive account of an existing phenomenon. The problem being addressed, namely the absence of an integrated, Taglish-aware, multimodal forensic tool for social media users, particularly in low-resource and code-switching linguistic contexts, is not one that can be resolved through observation or survey alone. It demands the actual construction and empirical evaluation of a novel technological artifact, which is precisely what the developmental design enables.

The developmental orientation of this research is further justified by the nature of the technical components being integrated. The system fuses a fine-tuned Filipino BERT model for text detection, a CNN-Vision Transformer hybrid for AI-generated image classification, and a frame-level forensic engine for deepfake video identification, each of which must be individually developed, trained, and verified before being combined into the unified late fusion architecture that constitutes PRISM as a whole. Cruz et al. (2020) demonstrated that the development of a benchmark Filipino fake news dataset and the fine-tuning of transfer learning models on that dataset are inherently developmental activities, as they produce functional outputs that did not previously exist for the target linguistic context. Similarly, Visweswaran et al. (2024)

and Jing et al. (2023) conducted developmental research when they implemented and evaluated their respective multimodal fusion systems on real-world datasets. The present study follows this established developmental tradition while extending it to the Taglish environment and the broader harm profile documented by Blancaflor et al. (2023, 2024) and Schipper (2025) in the Philippine context.

Developmental research of this scope requires a structured yet iterative process that can accommodate the discovery of technical limitations, the refinement of model architectures, and the incorporation of user feedback at meaningful checkpoints throughout the system's construction. For this reason, the present study adopts the Prototyping Model as its specific System Development Life Cycle.

The Prototyping Model, which has been a classical reference in software engineering literature since its foundational description in Pressman's *Software Engineering: A Practitioner's Approach*, provides a disciplined framework of iterative prototype construction, stakeholder evaluation, and systematic refinement that is uniquely suited to the AI-driven, user-facing nature of PRISM. The combination of a developmental research design with the prototyping development model ensures that the study produces a system that is both technically validated through empirical performance metrics and practically validated through direct engagement with social media users as the intended beneficiaries of the tool.

Research Approach

This study will employ a quantitative research approach as its primary mode of data treatment and analysis. The quantitative approach is appropriate for this research because the central questions of the study, concerning detection accuracy, precision, recall, F1-score, inference latency, and user experience scores, are all expressed in numerical terms that require statistical measurement and comparison rather than

interpretive or descriptive analysis. Jha (2025) similarly employed quantitative experimentation when reporting the 18 percent improvement in detection accuracy achieved by the multimodal framework over traditional systems on real-world benchmarks, and Alonso-Bartolome and Segura-Bedmar (2022) quantified the superiority of their multimodal CNN approach by reporting a statistically meaningful 9-percentage-point accuracy gain over text-only BERT baselines on the Fakeddit dataset. The present study adopts the same empirical framework for evaluating PRISM against single-modality detection baselines.

The quantitative approach operates across two distinct evaluation tracks within this study. The first track is technical performance evaluation, in which the detection accuracy, precision, recall, and F1-score of each of PRISM's three forensic modules will be measured using held-out test sets drawn from verified disinformation repositories including Vera Files, AFP Philippines, and established benchmark datasets such as FakeNewsNet. A Confusion Matrix will be used as the primary statistical tool for computing these metrics by comparing predicted classification labels against verified ground-truth labels for each piece of test content. This approach is consistent with the evaluation methodology used by Heidari et al. (2023) and Park et al. (2024), both of whom applied standard classification metrics to evaluate deepfake and AI-image detection systems under controlled experimental conditions. The multimodal results of the integrated PRISM system are then compared statistically against single-modality baselines to test the null hypothesis that no significant improvement in detection accuracy will be achieved by the multimodal architecture.

The second track is user experience evaluation, which will be quantified through the System Usability Scale (SUS), a validated ten-item psychometric instrument that produces a composite score from 0 to 100 reflecting the perceived usability of a system. The SUS has been widely adopted in human-computer interaction research because it is reliable, technology-agnostic, and sufficiently

sensitive to detect meaningful differences in user experience across system variants. Applying the SUS to a sample of social media users who interact with the PRISM prototype generates interval-scale data that can be analyzed through standard descriptive and inferential statistics, enabling the study to quantitatively assess whether the system's Explainable AI interface meets the usability thresholds required for practical deployment in the social media context documented by Valdez et al. (2021) and Szczepanski et al. (2021).

Research Settings

The study will be conducted in a research environment that serves as the primary institutional setting for all phases of the research, from system development and model training to prototype evaluation and user testing. This environment is particularly appropriate as research setting for this study because it concentrates a population of active social media users who are simultaneously students and educators, a demographic that Valdez et al. (2021) identified as both vulnerable to disinformation and capable of developing media literacy skills when provided with structured tools for credibility evaluation. The research environment provides access to the computing infrastructure required for prototype development, including laboratory workstations compatible with the hardware requirements specified for PRISM's model training and local inference tasks.

The digital setting of the study extends beyond the physical research environment to encompass the online platforms in which PRISM operates, specifically Facebook, TikTok, and X (formerly Twitter), which constitute the primary vectors for the spread of synthetic media and disinformation documented by Ong and Cabañes (2018) and Schipper (2025). User evaluation sessions will be conducted with participants accessing both the PRISM browser extension and the React web-based manual

submission interface in a controlled but naturalistic setting that replicates their ordinary social media browsing behavior, ensuring the SUS scores obtained are ecologically valid indicators of the system's practical effectiveness for its intended users.

Additionally, the cloud-based GPU environments used during the intensive model training phases of PRISM's development constitute a secondary computational setting for this research. Given the computational demands of fine-tuning transformer-based architectures such as Filipino BERT and the CNN-ViT hybrid on large multimodal datasets, local hardware alone will be supplemented with cloud GPU access where necessary to ensure sufficient training throughput. However, all prototype evaluation, latency stress testing, and user acceptance testing will be conducted on consumer-grade hardware to ensure that performance measurements obtained are representative of the actual device capabilities available to the target user population.

Participants of the Study

The participants of this study will be drawn from the general population of active social media users. For the user evaluation phase, the study will target social media users who regularly engage with content on Facebook, TikTok, or X, as these are the platforms for which PRISM is designed. Purposive sampling will be employed to ensure that participants meet the inclusion criteria of being active social media users with sufficient exposure to digital content to meaningfully evaluate the system's Explainable AI interface, consistent with the user profile identified by Blancaflor et al. (2023) in their survey of Filipino deepfake awareness. The study will target a minimum of thirty participants for the SUS evaluation, a sample size that has been established in the human-computer interaction literature as sufficient for detecting meaningful differences in usability scores with acceptable statistical power (Tullis & Stetson, 2004).

Participants will be recruited voluntarily, with informed consent obtained from each individual prior to their interaction with the prototype.

The data sources for the technical evaluation component of the study will be drawn from both local and international repositories of verified disinformation content. Locally, fact-checked articles and verified false claims will be sourced from Vera Files and AFP Philippines, the two primary fact-checking organizations operating within the Philippine media landscape. These local sources are essential for ensuring that PRISM's training data reflects actual disinformation narratives, code-switching patterns, and visual manipulation techniques that circulate within social media, rather than relying exclusively on foreign datasets that do not capture the linguistic and cultural specificity of Taglish content. Cruz et al. (2020) underscored the critical importance of locally curated datasets for Filipino fake news detection, demonstrating that models trained or fine-tuned on Philippine-specific data significantly outperform those relying solely on general English-language corpora.

Internationally sourced benchmark datasets supplement the local data across each of PRISM's three forensic modules. For the text detection module, the Fake News Filipino dataset developed by Cruz et al. (2020) serves as the primary evaluation benchmark. For the image detection module, datasets encompassing GAN-generated and diffusion-generated images will be used, informed by the dataset construction principles established by Baraheem and Nguyen (2023) and Park et al. (2024). For the deepfake video detection module, established benchmarks will be used to supplement locally collected social media video samples, with the dataset curation strategy informed by the quality and diversity requirements identified by the survey of Akhtar et al. (2024). All data sources will be pre-processed through cleaning, Taglish normalization, class balancing, and frame-level extraction as appropriate to each module before being used for model training or evaluation.

Research Instruments

The primary research instrument for the user evaluation phase of this study is the System Usability Scale (SUS), a standardized ten-item questionnaire developed by Brooke (1996) that measures the perceived usability of a system through a set of alternating positively and negatively worded statements rated on a five-point Likert scale. The SUS produces a composite score from 0 to 100, with scores above 68 considered above average and scores above 80 considered good usability. The SUS will be selected as the primary user evaluation instrument because of its reliability, its validity across a wide range of product types and user populations, and its established application in similar AI-assisted information tool evaluations. Participants will be asked to interact with a working prototype of the PRISM browser extension before completing the SUS questionnaire, ensuring that their responses reflect actual system interaction rather than hypothetical impressions.

For the technical evaluation of the system's detection modules, the Confusion Matrix serves as the primary analytical instrument, producing the true positive, true negative, false positive, and false negative counts from which Detection Accuracy, Precision, Recall, and F1-score will be computed. The Confusion Matrix is universally employed in the machine learning literature for evaluating binary and multi-class classification systems and is the standard instrument used in comparable detection studies by Heidari et al. (2023), Baraheem and Nguyen (2023), and Zhang et al. (2023). For latency evaluation, a custom-designed latency stress test protocol will be employed, which measures the average inference time in milliseconds from the moment a social media post will be detected by the browser extension to the moment the Explainable AI verdict will be displayed to the user, under simulated conditions of active social media scrolling on consumer-grade desktop hardware.

A researcher-developed pre-interaction demographic questionnaire will be administered to all user evaluation participants prior to the SUS to capture relevant

background variables including age, educational level, frequency of social media use, prior exposure to disinformation, and self-assessed digital literacy level. These variables will be used for descriptive characterization of the participant sample and for exploratory analysis of whether subgroups of participants evaluate the system's usability differently, consistent with the findings of Blancaflor et al. (2023) that prior exposure to deepfake content correlates with awareness and evaluative capacity. All instruments will be reviewed for content validity by the research adviser and by faculty members with expertise in human-computer interaction and artificial intelligence prior to their deployment in the study.

Data Collection Procedure

Data collection in this study proceeds through two parallel tracks that correspond to the technical evaluation and user evaluation components of the research, both of which will be conducted in a sequential manner aligned with the iterative cycles of the Prototyping Model. The technical data collection track begins during the first prototype cycle with the assembly and preprocessing of training and evaluation datasets from Vera Files, AFP Philippines, the Fake News Filipino dataset, and the supplementary international benchmark datasets. Preprocessing steps for the text data include Taglish normalization, which involves standardizing code-switched Tagalog-English tokens to a consistent format compatible with the Filipino BERT model's tokenizer, removal of duplicate entries, and class balancing to ensure that real and fabricated content are equally represented in the training splits. For the image data, preprocessing includes resizing, augmentation through horizontal flipping and brightness variation, and the extraction of frequency domain features for the CNN-ViT hybrid's multi-level feature representation. For the video data, frame-level extraction will be performed at standardized intervals, followed by spatial artifact analysis of

individual frames and temporal consistency analysis across consecutive frame sequences.

Following each prototype iteration cycle, the trained detection modules will be evaluated against held-out test sets to record Accuracy, Precision, Recall, and F1-score values, which will be logged systematically in a performance tracking spreadsheet that allows the research team to monitor improvement across refinement cycles. Latency stress tests will be conducted by running the prototype browser extension on the designated consumer-grade desktop test hardware under simulated active browsing conditions, with inference time recorded in milliseconds for each modality separately and for the combined late fusion verdict. These measurements will be collected for a minimum of one hundred test instances per modality to ensure stable latency estimates that are not disproportionately influenced by single-instance outliers.

The user evaluation data collection track will be activated when the prototype reaches a stage of functional completeness sufficient for meaningful interaction. Participants who have consented to participation will be scheduled for individual or small-group evaluation sessions. At the start of each session, participants complete the demographic questionnaire. They will be given a standardized orientation to the PRISM browser extension, including a brief demonstration of the Explainable AI verdict interface, and are subsequently allowed to interact with the prototype freely for a minimum of fifteen minutes using their own social media accounts or researcher-provided sample content. Following the interaction period, participants complete the SUS questionnaire. All completed SUS questionnaires will be collected immediately at the conclusion of each session to minimize non-response, and all responses will be anonymized before being entered into the statistical analysis dataset. The entire data collection process follows the ethical safeguards detailed in Section H of this chapter.

Data Analysis Techniques

The technical performance data generated by the Confusion Matrix evaluation of each detection module will be analyzed through standard classification metric computations to yield Detection Accuracy, Precision, Recall, and F1-score values for each module independently and for the integrated late fusion system as a whole. F1-score will be treated as the definitive comparative metric for the multimodal versus single-modality comparison because it provides a balanced view of both precision and recall, preventing a scenario where a model achieves deceptively high accuracy by systematically biasing toward the majority class in imbalanced test sets. This metric prioritization is consistent with the approach adopted by Jing et al. (2023), who used F1-score as the primary performance indicator for comparing the MPFN against prior state-of-the-art methods on the Twitter and Weibo datasets. The multimodal PRISM results will be compared against single-modality baselines for text-only detection, image-only detection, and video-only detection to directly test the hypothesis that the integrated system achieves significantly higher performance than any single-modality approach alone.

Latency data collected from the stress testing protocol will be analyzed through descriptive statistics including mean, median, and 95th percentile inference times per modality and for the combined verdict. The 95th percentile latency will be emphasized over the mean because it captures the worst-case performance experienced by users under typical browsing conditions and is therefore a more conservative and practically meaningful indicator of real-time viability than the average alone. This analysis approach will be informed by the real-time benchmarking methodology of Zhang et al. (2023), who similarly focused on processing time improvements under streaming conditions rather than simple average performance, and by the latency benchmarking of Tian et al. (2020), who demonstrated that fake news detection latency must be practically invisible to users during active scrolling.

User experience data collected through the SUS will be analyzed through the standard SUS scoring procedure, in which alternating positive and negative item scores will be transformed and summed to produce a composite score per participant, and descriptive statistics including mean, standard deviation, and percentile rank will be computed for the full sample. The resulting mean SUS score will be interpreted against the established SUS benchmarks to assess whether PRISM achieves the usability threshold required for practical deployment. Exploratory subgroup analysis will be conducted to examine whether SUS scores differ meaningfully across participant groups stratified by prior deepfake exposure, frequency of social media use, and self-assessed digital literacy level, consistent with the correlational findings of Blancaflor et al. (2023) regarding the relationship between social media exposure and disinformation awareness. All quantitative analyses will be performed using standard statistical software, and findings will be reported with appropriate measures of central tendency and dispersion.

Ethical Considerations

The ethical conduct of this research will be governed by the principles of respect for persons, beneficence, and justice as articulated in standard research ethics frameworks applicable to studies involving human participants and AI-generated data. Given that PRISM will be designed to process social media content and produce credibility verdicts about that content, the ethical implications of the system extend beyond the research process itself to encompass the deployment context in which the finished tool will operate. The study addresses ethical responsibilities at both levels.

With respect to human participants in the user evaluation phase, informed consent will be obtained from every participant prior to their enrollment in the study. The informed consent process communicates in plain language the purpose of the

study, the nature of the task's participants will perform, the types of data that will be collected, the anonymization procedures applied to all collected data, and the participant's right to withdraw from the study at any time without penalty. Participation is entirely voluntary, and no incentive structures will be employed that could constitute undue inducement. Demographic data and SUS responses will be stored in password-protected files accessible only to the research team, and all identifying information will be removed from the dataset before any analysis will be performed.

With respect to the data sources used for model training and evaluation, all content drawn from Vera Files, AFP Philippines, and publicly available benchmark datasets will be used in accordance with the terms of access established by each source. No private individual's personal data will be collected or used without consent. Social media content used for training will be drawn exclusively from publicly available posts that have been independently verified as either credible or fabricated by established fact-checking organizations, ensuring that the labeling of training data does not rely on the judgment of the research team alone. This approach is consistent with the data sourcing principles applied by Cruz et al. (2020) in the construction of the Fake News Filipino dataset, which was validated through expert curation rather than algorithmic labeling.

With respect to the design of the PRISM system itself, the ethical implications of deploying an AI-based credibility assessment tool are directly addressed through several architectural decisions. First, all verdicts produced by PRISM will be presented as probability scores and confidence-weighted signals rather than definitive binary classifications, explicitly framing the system as a decision-support tool rather than an authoritative arbiter of truth. This design decision guards against the cognitive offloading risk identified in the definition of Explainable AI within this study, wherein users may passively accept AI verdicts without independent critical evaluation, a concern that Valdez et al. (2021) flagged as a significant weakness in Filipino media

literacy contexts. Second, the system does not possess any content removal or censorship capability, ensuring that PRISM cannot be misused as a censorship instrument regardless of who operates it. Third, the use of Manifest V3 for the browser extension component, which was informed by the security analysis of Hassan et al. (2025), minimizes the data collection footprint of the extension and protects user privacy by restricting the permissions available to the system below what older extension architectures would permit. These ethical safeguards collectively ensure that PRISM will serve the verification and media literacy needs of social media users without introducing new risks of manipulation, censorship, or privacy violation into the environment it will be designed to protect.

System Architecture / Development Model

The development of PRISM will follow the Prototyping Model of the System Development Life Cycle (SDLC), as described in Pressman's *Software Engineering: A Practitioner's Approach* (Pressman, 2010), which is a seminal reference work that formally codified prototyping as a disciplined engineering methodology for software systems whose requirements are complex, partially known, or subject to change through user interaction. The Prototyping Model will be selected in preference to linear development approaches such as the Waterfall model because PRISM development requirements include several dimensions of inherent uncertainty that cannot be fully specified before construction begins. These dimensions will include the optimal fusion weights for combining text, image, and video module confidence scores into a unified late fusion verdict; the specific Taglish normalization parameters that will maximize Filipino BERT's classification performance on real-world social media data; the latency-accuracy tradeoffs achievable through DistilBERT-Tagalog compression under

real-time performance constraints; and the Explainable AI interface design that best communicates credibility signals to non-technical social media users without inducing the cognitive offloading risks documented by Valdez et al. (2021). None of these parameters can be definitively established through specification documents alone; they will only be discovered and validated through cycles of construction, testing, and refinement.

The Prototyping Model will structure PRISM's development as an iterative loop between four stages: Requirements Gathering, Quick Design, Prototype Construction, and Prototype Evaluation followed by Prototype Refinement. In the Requirements Gathering stage, the functional and non-functional requirements of PRISM will be documented based on the research questions established in the introduction, the technical capabilities demonstrated in the literature by Visweswaran et al. (2024), Jing et al. (2023), and Botnevik et al. (2020), and the user needs identified through the misinformation vulnerability analysis of Valdez et al. (2021) and Schipper (2025). The requirements will include real-time detection latency below two seconds per content unit, a minimum F1-score of 90 percent for the text and image modules on social media test content, a minimum SUS score of 68 for the Explainable AI interface, and full compatibility with the Manifest V3 browser extension standard. In the Quick Design stage, the architectural blueprint of PRISM will be drafted without exhaustive technical specification, producing a working conceptual design sufficient to guide the construction of the first prototype. This design will specify the late fusion architecture at a structural level, identifying the three independent forensic modules and the fusion mechanism through which their outputs will be combined. It will also specify the Supabase backend platform that will manage structured data and API routing, FastAPI for AI model inference serving, and the Explainable AI interface that will render forensic verdicts to users. The Quick Design stage will draw directly from the architectural precedents of BRENDA (Botnevik et al., 2020) for the browser extension delivery layer,

and the TextGCN-ViT fusion system of Visweswaran et al. (2024) for the multimodal integration architecture.

In the Prototype Construction stage, the Quick Design will be translated into a functional, deployable prototype of PRISM. The first prototype cycle will focus on establishing the core functionality of each forensic module in isolation before integration: the Filipino BERT text classifier will be fine-tuned on the Fake News Filipino dataset and local Vera Files content; the CNN-ViT image classifier will be initialized through transfer learning from pre-trained weights and fine-tuned on the combined local and benchmark image datasets; and the frame-level forensic engine will be constructed and tested on deepfake video benchmarks. Subsequent prototype cycles will integrate these modules through the late fusion mechanism and deploy them within the browser extension interface. The browser extension will be built using the Manifest V3 standard to ensure compliance with the security requirements identified by Hassan et al. (2025). Model compression through Knowledge Distillation to the DistilBERT-Tagalog variant will be applied during later prototype cycles to meet the real-time latency requirements established in the requirements gathering stage. In the Prototype Evaluation stage, the constructed prototype will be subjected to the quantitative technical and user experience evaluation procedures described in Sections F and G of this chapter. Technical evaluation will measure Detection Accuracy, Precision, Recall, F1-score, and inference latency against the established benchmarks, while user evaluation will measure SUS scores from social media users who will interact with the prototype. The evaluation findings will be analyzed and documented in a structured prototype evaluation report that will identify specific performance gaps or usability deficiencies requiring correction. This report will form the input to the Prototype Refinement stage, in which the identified gaps will be addressed through targeted adjustments to model hyperparameters, training data composition, fusion weight configurations, interface design elements, or system architecture as appropriate. The

refined system will then be submitted to another cycle of Prototype Construction and Prototype Evaluation, with the iterative loop continuing until all established performance and usability requirements will be met.

The full architecture of the accepted PRISM system will consist of four primary deployment components. The Supabase Backend Platform will serve as the central data and API management layer, providing PostgreSQL-based structured storage for detection logs and user session records, along with user authentication, real-time data subscriptions, and API routing through Supabase Edge Functions. Deployed separately, the FastAPI Service Layer will handle compute-intensive AI model inference, routing content inputs through the appropriate forensic modules and returning Explainable AI verdicts to the client interfaces. The Manifest V3 browser extension will operate as a passive content scanner within the user's browser and a React-based web application will also be developed to serve as a manual content submission interface, allowing users to directly upload or paste text, images, and video content for forensic analysis outside of the passive browser extension context, detecting social media posts as they load and submitting them to the FastAPI layer for forensic analysis and to Supabase for session management without requiring user-initiated action. Together, these three components will constitute an end-to-end real-time multimodal disinformation detection system that will embody the technical contributions, user-centered design principles, and ethical safeguards established throughout this methodology chapter.

REFERENCES

Alonso-Bartolome, S., & Segura-Bedmar, I. (2022). Multimodal fake news detection. *Information*, 13(6), Article 284. <https://doi.org/10.3390/info13060284>

- Akhtar, A., Kaur, A., Hoshyar, A. N., Saikrishna, V., Firmin, S., & Xia, F. (2024). Deepfake video detection: Challenges and opportunities. *Artificial Intelligence Review*, 57, Article 1. <https://doi.org/10.1007/s10462-024-10810-6>
- Baraheem, S. S., & Nguyen, T. V. (2023). AI vs. AI: Can AI detect AI-generated images? *Journal of Imaging*, 9(10), Article 199. <https://doi.org/10.3390/jimaging9100199>
- Blancaflor, E., Anave, D. M., Cunanan, T. S., Frias, T., & Velarde, L. N. (2023). A literature review of the legislation and regulation of deepfakes in the Philippines. In *Proceedings of the 2023 ACM International Conference* (pp. 1–8). ACM. <https://dl.acm.org/doi/abs/10.1145/3616712.3616722>
- Blancaflor, E., Garcia, J. I., Magno, F. D., & Vilar, M. J. (2024). Deepfake blackmailing on the rise: The burgeoning posterity of revenge pornography in the Philippines. In *Proceedings of the 2024 ACM International Conference* (pp. 1–8). ACM. <https://dl.acm.org/doi/abs/10.1145/3654522.3654548>
- Botnevik, B., Sakariassen, E., & Setty, V. (2020). BRENDA: Browser extension for fake news detection. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 2117–2120). ACM. <https://dl.acm.org/doi/abs/10.1145/3397271.3401396>
- Chuan, C., et al. (2024). Explainable Artificial Intelligence (Xai) for facilitating recognition of algorithmic bias: An experiment from imposed users' perspectives. *Journal of Artificial Intelligence and Human Rights*, 12(2), 45–62. <https://www.sciencedirect.com/science/article/pii/S073658532400039X>
- Cruz, J. C. B., Tan, J. A., & Cheng, C. (2020). Localization of fake news detection via multitask transfer learning. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)* (pp. 2596–2604). European Language Resources Association. <https://aclanthology.org/2020.lrec-1.316/>
- Cui, L., Wang, S., & Lee, D. (2019). SAME: Sentiment-aware multi-modal embedding for detecting fake news. In *Proceedings of the 2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)* (pp. 41–48). IEEE. <https://ieeexplore.ieee.org/document/9073233>
- Department of Information and Communications Technology. (2025, May 23). AI tool "Tunai" wins in inaugural OpenGov Hackathon. <https://dict.gov.ph/news-and-updates/21070>
- Diwan, A., Mittal, A. R., Khare, S., Bharadwaj, S., & Dhamecha, T. I. (2021). Multilingual, code-switching and low-resource NLP and ASR [Master's thesis, Indian Institute of Technology Bombay]. Semantic Scholar. <https://www.semanticscholar.org/paper/Multilingual%2C-Code-switching-and-Low-Resource-NLP-Diwan-Mittal/17f75b694792c94bfbbb4a0eafcd3d0866599ca3>
- Hassan, S. M. T. U., Zeeshan, A., & Ahmed, U. (2025). Insecure by design: A study of permission abuse and vulnerabilities in browser extensions. *Academia*. https://www.academia.edu/143364824/Insecure_By_Design_A_Study_Of_Permission_Abuse_and_Vulnerabilities_In_Browser_Extensions
- Heidari, A., Navimipour, N. J., Dag, H., & Unal, M. (2023). Deepfake detection using deep learning methods: A systematic and comprehensive review. *WIREs Data Mining and Knowledge Discovery*, 14(2), Article e1520. <https://doi.org/10.1002/widm.1520>
- Jha, A. (2025). How artificial intelligence can prevent social media fraud: A multimodal detection framework. *International Journal on Science and Technology*, 16(4). <https://doi.org/10.71097/IJSAT.v16.i4.9553>
- Jing, J., Wu, H., Sun, J., Fang, X., & Zhang, H. (2023). Multimodal fake news detection via progressive fusion networks. *Information Processing & Management*, 60(1), Article 103120. <https://doi.org/10.1016/j.ipm.2022.103120>
- Ong, J. C., & Cabañes, J. V. A. (2018). Architects of networked disinformation: Behind the scenes of troll accounts and fake news production in the Philippines. Newton Tech4Dev Network.

- <https://scholarworks.umass.edu/entities/publication/01d06f54-c7f2-4103-96c1-168a16f9028b>
- Park, D., Na, H., & Choi, D. (2024). Performance comparison and visualization of AI-generated-image detection methods. *IEEE Access*, 12, 1–12. <https://ieeexplore.ieee.org/abstract/document/10508937>
- Pathak, R., Spezzano, F., & Pera, M. S. (2023). Understanding the contribution of recommendation algorithms on misinformation recommendation and misinformation dissemination on social networks. *ACM Transactions on the Web*, 18(1), Article 1. <https://dl.acm.org/doi/full/10.1145/3616088>
- Schipper, T. (2025). Disinformation by design: Leveraging solutions to combat misinformation in the Philippines' 2025 election. *Data & Policy*, 7, Article e21. <https://doi.org/10.1017/dap.2025.3>
- Szczepański, M., Pawlicki, M., Kozik, R., & Choraś, M. (2021). New explainability method for BERT-based model in fake news detection. *Scientific Reports*, 11, Article 23705. <https://doi.org/10.1038/s41598-021-03100-6>
- Tian, L., Zhang, X., & Peng, M. (2020). FakeFinder: Twitter fake news detection on mobile. In *Companion Proceedings of the Web Conference 2020* (pp. 243–246). ACM. <https://dl.acm.org/doi/abs/10.1145/3366424.3382706>
- Tolstykh, I., Tsybina, A., Yakubson, S., Gordeev, A., Dokholyan, V., & Kuprashevich, M. (2024). GigaCheck: Detecting LLM-generated content via object-centric span localization. *arXiv*. <https://arxiv.org/abs/2410.23728>
- Valdez, P. N., Ugalingan, G., & Garinto, L. A. (2021). Challenging misinformation in the online English language classroom in the Philippines. *MEXTESOL Journal*, 45(3). https://www.mextesol.net/journal/index.php?page=journal&id_article=46333
- Visweswaran, M., Mohan, J., Kumar, S. S., & Soman, K. P. (2024). Synergistic detection of multimodal fake news leveraging TextGCN and vision transformer. *Procedia Computer Science*, 235, 142–151. <https://doi.org/10.1016/j.procs.2024.04.017>
- Zhang, C., Gupta, A., Qin, X., & Zhou, Y. (2023). A computational approach for real-time detection of fake news. *Expert Systems with Applications*, 216, Article 119458. <https://doi.org/10.1016/j.eswa.2023.119458>